

Problem 3.1 (MATLAB)

Hypothesis test and confidence region/intervals for one-sample multivariate normal observations

The file '**dataset_problem_3_1.mat**' contains the 30x2 data matrix for a sample of 30 pieces of a particular grade of lumber. The first column (x_1) contains measurements of stiffness (modulus of elasticity) and the second column (x_2) contains the corresponding measurements of bending strength for the lumber pieces, both values in units proportional to $[N/m^2]$.

It is desired to test if the sample can be said to live up to a particular standard with respect to these two parameters.

Assuming first, that $(X_1, X_2) \sim N_2(\mu, \Sigma)$ with μ, Σ both unknown

- Load the data matrix
- Plot the observations (scatter plot)
- Calculate the descriptive statistics $\hat{\mu}, \hat{\Sigma}$
- Perform an exact test of the hypothesis, that the lumber pieces fulfills the relevant standard, that is, test $H_0: \mu = \mu_0 = [1750 \ 950]^T$ against $H_1: \mu \neq \mu_0$. Use level of significance $\alpha = 0.05$ but calculate also the p-value for the test
- Perform an large-sample approximate test of the same hypothesis and use again $\alpha = 0.05$ but calculate also the p-value
- Compare the two test-results and conclude

Turning to confidence region and intervals, still using $\alpha = 0.05$,

- Calculate (and optionally plot) the $100(1 - \alpha)\%$ confidence region for (μ_1, μ_2) . Is the targeted (2D) mean μ_0 inside the confidence region ?
- Calculate simultaneous $100(1 - \alpha)\%$ confidence intervals for μ_1 and μ_2
- Calculate marginal (one-at-a-time) $100(1 - \alpha)\%$ confidence intervals for μ_1 and μ_2
- Calculate Bonferroni $100(1 - \alpha)\%$ confidence intervals for μ_1 and μ_2
- Compare the various confidence intervals and explain
- Also compare with the hypothesis test results

Finally, perform a model check for multivariate normality:

- Make plots of the multivariate scatter matrix plus marginal histograms and boxplots
- Calculate the squared sample Mahalanobis distances, $d_i^2 = (x_i - \bar{x})^T S^{-1} (x_i - \bar{x})$, and make a QQ-plot versus the relevant χ^2 distribution for check of approximate linearity
- Conclude

Problem 3.2 (Theoretical)

Connection between univariate Student-t and F distributions

Show that

$$X \sim t(n-1) \Rightarrow X^2 \sim F(1, n-1)$$

thus giving the connection between the univariate standard t-test and the special case $p=1$ for the multivariate Hotelling T^2 -test.

Hint: For an easy argument, refer to the definitions of Student-t and F distributed random variables in terms of standard normal and chi-squared distributed random variables.

Problem 3.3 (Theoretical + optionally MATLAB)

One-sample Likelihood Ratio Test for multivariate covariance matrix

For a multivariate normal distribution

$$X \sim N_p(\mu, \Sigma)$$

with μ, Σ both unknown and the observations

$$X_1, X_2, \dots, X_n \text{ iid } \sim N_p(\mu, \Sigma)$$

it can be shown, that the Likelihood Ratio test-statistic, Λ , for a test of a specific (matrix) value for the unknown covariance matrix, that is, the test

$$H_0: \Sigma = \Sigma_0 \text{ against } H_1: \Sigma \neq \Sigma_0,$$

can be expressed as

$$-2 \log \Lambda = -2 \log \frac{\max_{\mu} L(\mu, \Sigma_0)}{\max_{\mu, \Sigma} L(\mu, \Sigma)} = n \text{ trace}(\Sigma_0^{-1} S) - n \log |\Sigma_0^{-1} S| - np$$

where S is the usual sample covariance matrix estimator.

Device a large-sample approximate test for H_0 by considering the degrees of freedom (number of unrestricted scalar parameters) for the nominator and denominator maximizations above, respectively.

Optionally you can try to “test your test” in MATLAB by defining μ, Σ , generate observations and perform your test in situations where H_0 is true and where H_1 is true, respectively

Problem 3.4 (Theoretical + optionally MATLAB) **Derivation of Likelihood Ratio test-statistics (examples)**

Part a: LRT for univariate Poisson distribution

Consider a univariate Poisson distributed random variable

$$X \sim \text{Poisson}(\lambda) \Rightarrow f_X(x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, 2, \dots$$

and the iid observations $x_1, x_2, \dots, x_n$.

Give an expression for the likelihood function

$$L(\text{obs}|\lambda) = L(x_1, x_2, \dots, x_n|\lambda)$$

Find the maximum likelihood estimator for λ , $\hat{\lambda}_{ML}$, by differentiating L or equivalently $\log L$.

Next, for an Likelihood Ratio test of

$$H_0: \lambda = \lambda_0 \text{ against } H_1: \lambda \neq \lambda_0,$$

find an expression for the likelihood ratio

$$\Lambda = \frac{L(\text{obs}|\lambda_0)}{\max_{\lambda} L(\text{obs}|\lambda)}$$

using in the denominator the derived maximizing argument $\lambda = \hat{\lambda}_{ML}$.

Finally use this to device a large-sample approximate LR-test of H_0 (using $-2\log \Lambda$) by considering degrees of freedom for the nominator and denominator of Λ .

Optionally you can try to “test your test” in MATLAB by defining λ , generate observations and perform your test in situations where H_0 is true and where H_1 is true, respectively

Part b: LRT for univariate exponential distribution

Consider a univariate exponentially distributed random variable

$$T \sim \text{Exp}(\lambda) \Rightarrow f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0$$

and the iid observations $t_1, t_2, \dots, t_n$.

Give an expression for the likelihood function

$$L(\text{obs}|\lambda) = L(t_1, t_2, \dots, t_n|\lambda)$$

Find the maximum likelihood estimator for λ , $\hat{\lambda}_{ML}$, by differentiating L or equivalently $\log L$.

Next, for an Likelihood Ratio test of

$$H_0: \lambda = \lambda_0 \text{ against } H_1: \lambda \neq \lambda_0,$$

find an expression for the likelihood ratio

$$\Lambda = \frac{L(\text{obs}|\lambda_0)}{\max_{\lambda} L(\text{obs}|\lambda)}$$

using in the denominator the derived maximizing argument $\lambda = \hat{\lambda}_{ML}$.

Finally use this to device a large-sample approximate LR-test of H_0 (using $-2\log \Lambda$) by considering degrees of freedom for the nominator and denominator of Λ .

Optionally you can try to “test your test” in MATLAB by defining λ , generate observations and perform your test in situations where H_0 is true and where H_1 is true, respectively